

IC 250: Programming and Data Structure Practicum

Course Code	IC 250
Course Name	Programming and Data Structure Practicum
L-T-P-C	1-0-3-3
Prerequisites	IC 150
Intended for	B.Tech 1st and 2nd
Distribution	Core
One-time approval	2nd Adhoc Senate (May 2012), in the B.Tech Institute Core (common core courses) curriculum; recommended by the 9th BoA (Oct 2015)
Discontinued	17th Senate (Aug 2018)

Course Contents

Introduction to data structures, abstract data types, Creation and manipulation of data structures: arrays, lists, stacks, queues, trees, heaps, hash tables, balanced trees. Algorithms for sorting and searching. Notion of time and space complexity, the O-notation

- **Introduction:** Role of algorithms in Computing, analyzing algorithms and designing algorithms [2 Lectures]
- **Data Structures:** Stacks, queues, linked lists, rooted trees, B-tree, graphs, hash tables, recursion [6 Lectures]
- **Sorting and Searching Algorithms:** Bubble, Heapsort, Quicksort, Sequential Searches, Binary search, [4 Lectures]
- **Efficiency:** Time and Space complexity, O-Notation, Space time trade-off, Measuring execution time, memory usage [2 Lectures]

Laboratory Modules

Lab to be conducted on a 3-hour slot. It will be conducted in tandem with the theory course so the topics for problems given in the olab are already initiated in the theory cloass. The topics taught in the theory course should be approximately be sequenced for synchronization with the laboratory. A sample sequence of topics and lab classes for the topic are given below

- Two assignments: Designing algorithm for some problems and writing program for it
- Four-Five assignments (some examples: Building a queue of strings, practice with linked data structures, Using a stack to evaluate arithmetic expressions etc),
- Two-Three assignments: sorting with recursion etc.
- Four-Five assignments: COmparing time and space complexity, e.g., comparing sorting by minimum search and sorting by mergesort, Analyzing NP hard and NP complete problems and dealing with them.

Text Books

1. Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein, **Introduction to Algorithms**, MIT Press

References

1. Alfred V. Aho, Jeffrey D. Ullman, John E. Hopcroft, **Data Structures and Algorithms**, Pearson.